Mohsin SHAIKH

AI Engineer | LLM Systems, AI Agents, RAG & Production ML

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Location: Berlin, Germany **Portfolio:** randomMatrix77.github.io



PROFILE

AI Engineer with ~3 years building production LLM systems in a regulated domain (insurance and financial advice), including AI systems that generate and validate software workflows, and 5+ years of applied ML overall across RAG, AI agents, NLP, document AI, and Python/FastAPI services on Kubernetes.

Selected achievement: Co-author of the first-place solution for both ISIC 2019 Skin Lesion Classification tasks.

SKILLS

Applied AI: LLM applications, RAG, AI agents/multi-agent systems, DSPy, LangGraph, LangChain, Recursive Language Models, Azure OpenAI Realtime, multimodal/VLM, LLM and RAG evaluation (ragas), rerankers, guardrails, Microsoft Decision-1
Machine Learning: PyTorch, Hugging Face, fine-tuning, LoRA, quantisation, DeepSpeed, NLP, NER/NEL, OCR, computer vision
Software & Data: Python, TypeScript, SQL, FastAPI, WebSockets/streaming APIs, PostgreSQL, SQLAlchemy/Alembic, Supabase, Redis/Valkey, Kafka, Milvus, pgvector, Azure AI Search, ChromaDB, FAISS, Z3 SMT solver
Agent Tooling: MCP, GitHub Copilot cloud agents, Claude Code, Codex, OpenAI APIs, Google ADK
Cloud & MLOps: Azure (AKS, AI Foundry), AWS, Docker, Kubernetes, Helm, Kubeflow, GitHub Actions, Keycloak/SSO
Languages: English (C2), German (B1)

EXPERIENCE

Machine Learning Engineer | Applied AI & LLM Systems — October 2023–Present

muffintech GmbH — *Berlin, Germany*

Start-up building conversational AI for the insurance and financial-advisory industry (Germany & Austria).

- Built a multi-stage LLM pipeline (requirements → scenarios → modules → implementation) that turns plain-language requests into runnable multi-agent workflows, Z3-validated before publishing, and edits live ones from change requests; now building a DSPy ReAct agent that reuses precedent workflows and verifies builds by replaying and judging traces.
- Integrated existing text workflows into an Azure OpenAI Realtime voice agent with session-owned tools, interrupt handling, background-noise filtering, voice-specific guardrails, and timed voice exams, delivering a production voice mode.
- Doubled top-3 recall in internal testing with three-stage hierarchical RAG and a pre-retrieval document resolver; migrated retrieval from embedded and self-hosted Chroma to Milvus and Azure vector stores/pgvector.
- Developed a Recursive Language Model pipeline for non-indexed document collections (95% accuracy on an internal test set) and versioned knowledge updates with tenant isolation, controlled repointing, and rollback.
- Built guardrails for a regulated domain: Microsoft Decision-1 for citation/source verification, a hybrid citation verifier and source-relevance judge, scope/jailbreak checks, and a default crisis guardrail with region-specific support texts.
- Onboarded new enterprise tenants end to end across Azure, Kubernetes, and all FastAPI/PostgreSQL services on AKS; secured releases with pinned module versions, CI end-to-end tests, and QA pass-rate checks against the live version.
- Built an MCP server (FastAPI, Docker, Kubernetes) that sends Linear sub-issues to GitHub Copilot cloud agents, which open PRs in the target repos, with task/PR status reporting, enabling agent-driven cross-repo code changes.

R&D Engineer | Machine Learning, NLP & Document AI — May 2021–September 2023

STTech GmbH — *Munich, Germany*

- Fine-tuned spaCy, BERT-large, and RoBERTa NER/NEL models on custom clinical corpora and automated their training and zero-downtime release through Kubeflow.
- Automated insurance-document ingestion with a CNN-based layout classifier and tuned Tesseract OCR pipeline, reducing extraction errors on complex forms by approximately 40%.
- Fine-tuned and served LLMs on multi-GPU AWS instances (DeepSpeed, Colossal-AI, Accelerate) and built LangChain agents for document Q&A extraction (2022–23); cut GPU cost by about 30% and memory by 3x with LoRA and quantisation.
- Developed Kafka/FastAPI production services and an LLM-assisted Robot Framework tool that converted plain-English scenarios into automated tests for microservices.

Personal Projects

Mosaic | On-Device Multimodal AI 2026
github.com/randomMatrix77/mosaic *Kotlin, Python, llama.cpp, OpenCL, SQLite*

- Built an Android proof of concept running Qwen 3.5 4B text and image inference locally through llama.cpp, with OpenCL GPU offloading verified on a Galaxy Z Fold 6 and a Jetpack Compose interface connected to Python 3.11 through Chaquopy and JNI.
- Implemented EmbeddingGemma 2 text/image embeddings, semantic image retrieval, SQLite persistence, and evidence-backed contextual memory; added a bounded, allow-listed tool loop for memory, image, and web retrieval while keeping final Qwen reasoning on-device.

Autonomous UI Replication Agent 2026
github.com/randomMatrix77/ui_copycat *Python, LangChain, Gemini, Playwright, React*

- Built an observe–generate–evaluate proof of concept that captures live UI structure, styles, and interaction states, then generates React/Tailwind replicas using selectively queried DOM/CSS evidence.
- Combined deterministic DOM diffs with multimodal visual evaluation to produce targeted refinement feedback under bounded tool-call and iteration budgets.

Zero-Shot Computer-Use Agent 2026
github.com/randomMatrix77/infinite-craft-computer-use *Python, Gemma 4, LangChain, PyAutoGUI*

- Built a DOM-independent agent that plays Infinite Craft using a locally hosted Gemma 4 26B vision-language model for visual grounding, inventory reading, planning, and coordinate localisation.
- Implemented a custom stateful planning loop with action history, step budgets, hallucination checks, and deterministic PyAutoGUI execution.

Earlier Experience & Research

Working Student | ML Annotation Tooling June 2020–August 2020
Hamburg University of Technology (TUHH) *Hamburg, Germany*

- Built and packaged a cross-platform PyQt5 annotation tool with box/polygon labelling, YOLO export, and frame-accurate audio inspection for deep-learning datasets.

Research Assistant & Master's Thesis July 2018–June 2019
Institute of Computational Neuroscience (UKE) *Hamburg, Germany*

- Co-authored the first-place solution for both ISIC 2019 Skin Lesion Classification tasks, combining multi-resolution deep-learning ensembles with patient metadata.
- Built PyTorch pipelines for 3-D MRI segmentation, including a 7th-place MRBrainS18 residual U-Net, and benchmarked biologically plausible alternatives to backpropagation for the master's thesis.

Education

M.Sc. in Mechatronics 2016–2020
Hamburg University of Technology (TUHH) *Hamburg, Germany*

Thesis: *Scalability of Bio-Inspired Credit Assignment Techniques for Deep Learning in Medical Imaging*

B.E. in Mechanical Engineering 2011–2015
Gujarat Technological University *Navsari, India*

Publication

N. Gessert, M. Nielsen, **M. Shaikh**, R. Werner, and A. Schlaefer. "Skin Lesion Classification Using Ensembles of Multi-Resolution EfficientNets with Meta Data." *MethodsX*, 7:100864, 2020. doi:10.1016/j.mex.2020.100864